Day: 17 (04-11-2026)	**Multiple dataset ingestion and transformation in Snowflake**	Documented By:
Task – 2		Sujit T K

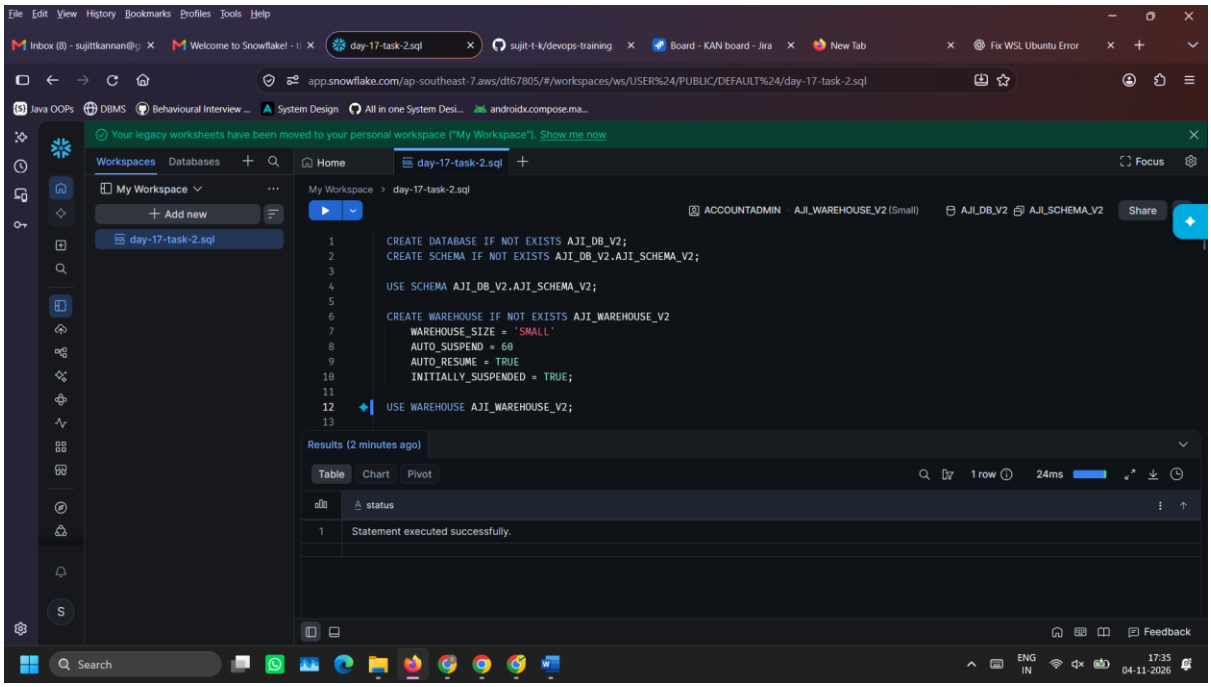
1.) Setup database, schema and warehouse

```
CREATE DATABASE IF NOT EXISTS AJI_DB_V2;
CREATE SCHEMA IF NOT EXISTS AJI_DB_V2.AJI_SCHEMA_V2;

USE SCHEMA AJI_DB_V2.AJI_SCHEMA_V2;

CREATE WAREHOUSE IF NOT EXISTS AJI_WAREHOUSE_V2
    WAREHOUSE_SIZE = 'SMALL'
    AUTO_SUSPEND = 60
    AUTO_RESUME = TRUE
    INITIALLY_SUSPENDED = TRUE;

USE WAREHOUSE AJI_WAREHOUSE_V2;
```

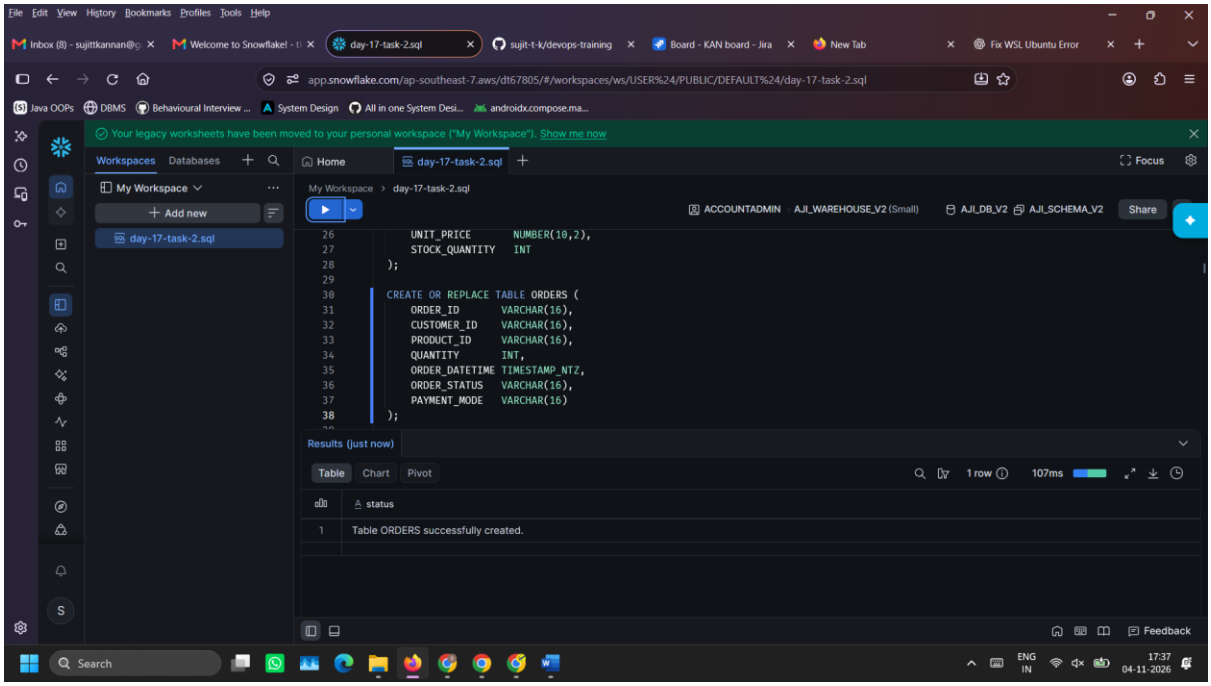


2.) Create tables: Customers, Products, Orders (to hold data)

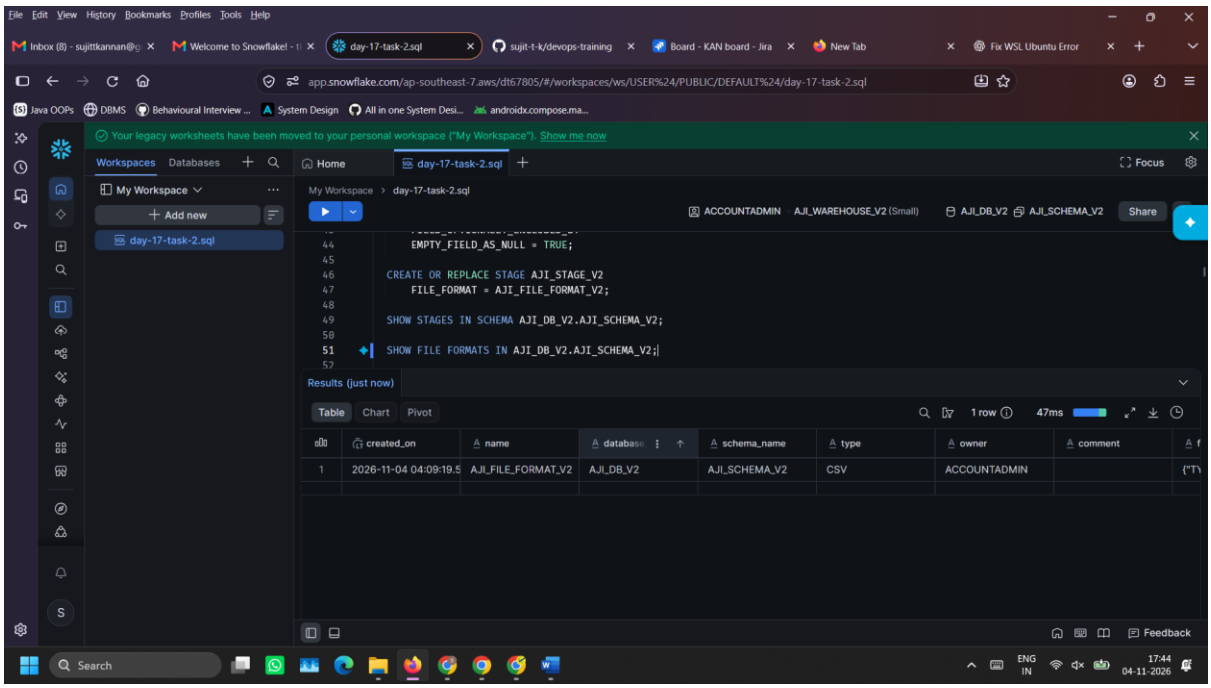
```
CREATE OR REPLACE TABLE CUSTOMERS (
    CUSTOMER_ID   VARCHAR(16),
    CUSTOMER_NAME VARCHAR(64),
    CITY          VARCHAR(32),
    SIGNUP_DATE   DATE,
    CUSTOMER_TIER VARCHAR(16)
);

CREATE OR REPLACE TABLE PRODUCTS (
    PRODUCT_ID       VARCHAR(16),
    PRODUCT_NAME     VARCHAR(64),
    PRODUCT_CATEGORY VARCHAR(32),
    UNIT_PRICE       NUMBER(10,2),
    STOCK_QUANTITY   INT
);
```

```
CREATE OR REPLACE TABLE ORDERS (
    ORDER_ID        VARCHAR(16),
    CUSTOMER_ID     VARCHAR(16),
    PRODUCT_ID      VARCHAR(16),
    QUANTITY        INT,
    ORDER_DATETIME  TIMESTAMP_NTZ,
    ORDER_STATUS    VARCHAR(16),
    PAYMENT_MODE    VARCHAR(16)
);
```



3.) Create file format and stage (so that the files can be uploaded into snowflake environment)




```
CREATE OR REPLACE FILE FORMAT AJI_FILE_FORMAT_V2
    TYPE = CSV
```

```
    SKIP_HEADER = 1
    FIELD_OPTIONALLY_ENCLOSED_BY = '"'
    EMPTY_FIELD_AS_NULL = TRUE;

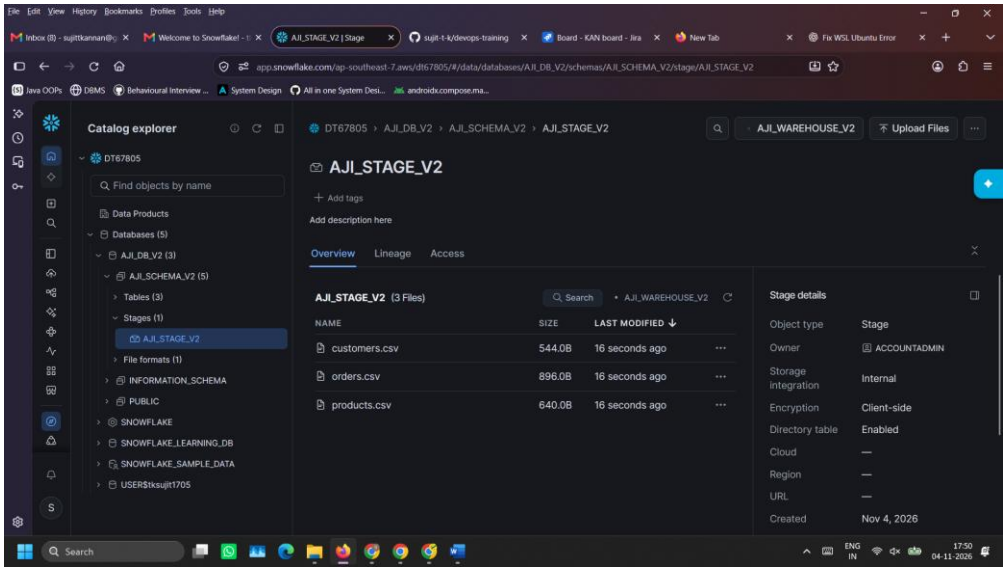
CREATE OR REPLACE STAGE AJI_STAGE_V2
    FILE_FORMAT = AJI_FILE_FORMAT_V2;

SHOW STAGES IN SCHEMA AJI_DB_V2.AJI_SCHEMA_V2;

SHOW FILE FORMATS IN AJI_DB_V2.AJI_SCHEMA_V2;
```

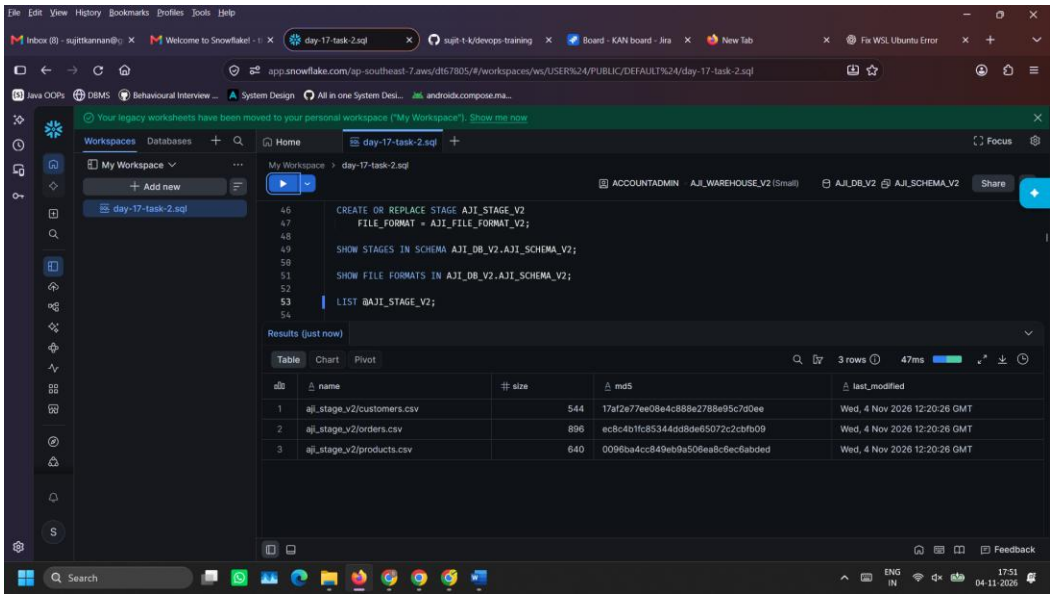
4.) Upload .csv files into the stage

- Click on ‘Catalog’ tab on the left sidebar and then go to Explorer -> AJI_DB_V2 -> AJI_SCHEMA_V2 -> Stages -> AJI_STAGE_V2.
- Then on the top right, you can find ‘Upload Files’ button. Click on it, a file chooser will open. Upload all files and wait till it gets uploaded.



5.) Verify files upload into the stage using SQL

```
LIST @DB_AJI_V2.AJI_STAGE_V2;
```

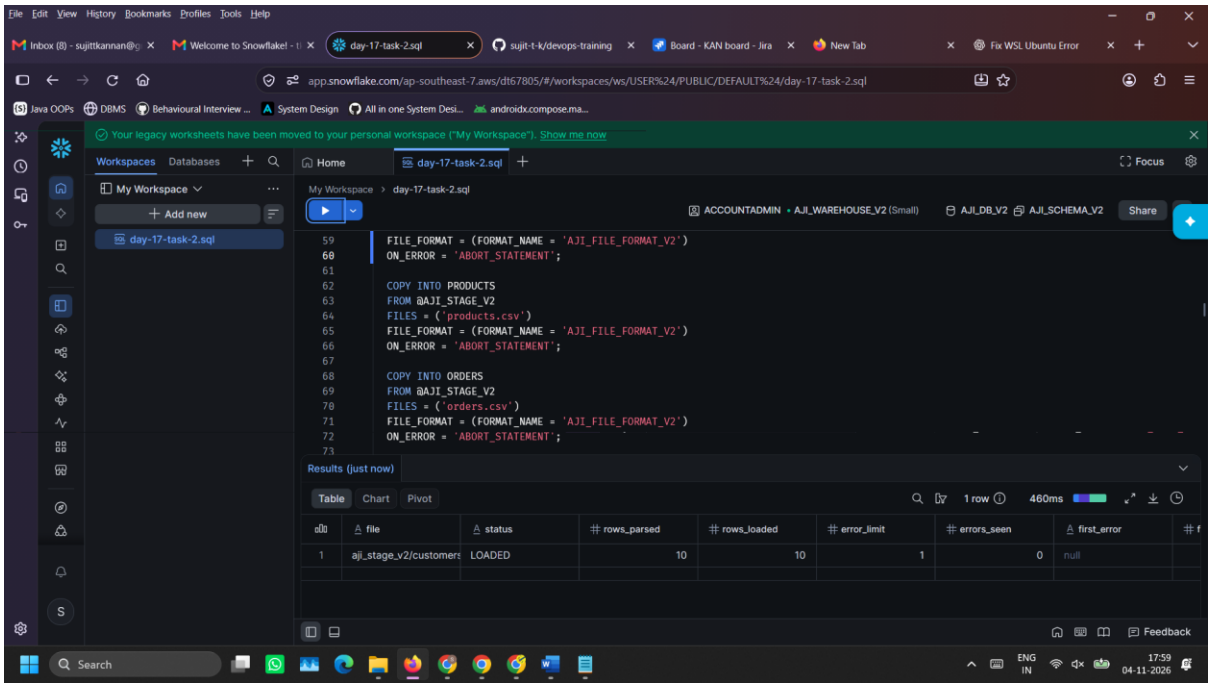


6.) Load data into tables using COPY INTO command

```
COPY INTO CUSTOMERS
FROM @AJI_STAGE_V2
FILES = ('customers.csv')
FILE_FORMAT = (FORMAT_NAME = 'AJI_FILE_FORMAT_V2')
ON_ERROR = 'ABORT_STATEMENT';

COPY INTO PRODUCTS
FROM @AJI_STAGE_V2
FILES = ('products.csv')
FILE_FORMAT = (FORMAT_NAME = 'AJI_FILE_FORMAT_V2')
ON_ERROR = 'ABORT_STATEMENT';

COPY INTO ORDERS
FROM @AJI_STAGE_V2
FILES = ('orders.csv')
FILE_FORMAT = (FORMAT_NAME = 'AJI_FILE_FORMAT_V2')
ON_ERROR = 'ABORT_STATEMENT';
```



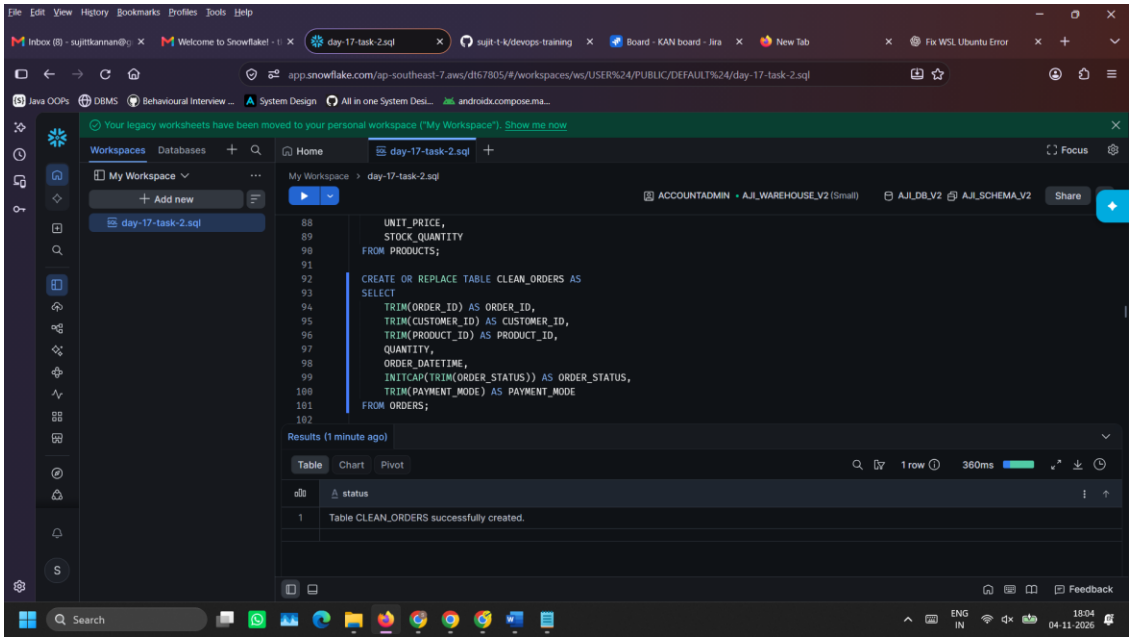
7.) Transform data (data cleaning) and save it in new table

```
CREATE OR REPLACE TABLE CLEAN_CUSTOMERS AS
SELECT
    TRIM(CUSTOMER_ID) AS CUSTOMER_ID,
    TRIM(COALESCE(CUSTOMER_NAME, 'UNKNOWN CUSTOMER')) AS CUSTOMER_NAME,
    UPPER(TRIM(CITY)) AS CITY,
    SIGNUP_DATE,
    UPPER(TRIM(CUSTOMER_TIER)) AS CUSTOMER_TIER
FROM CUSTOMERS;

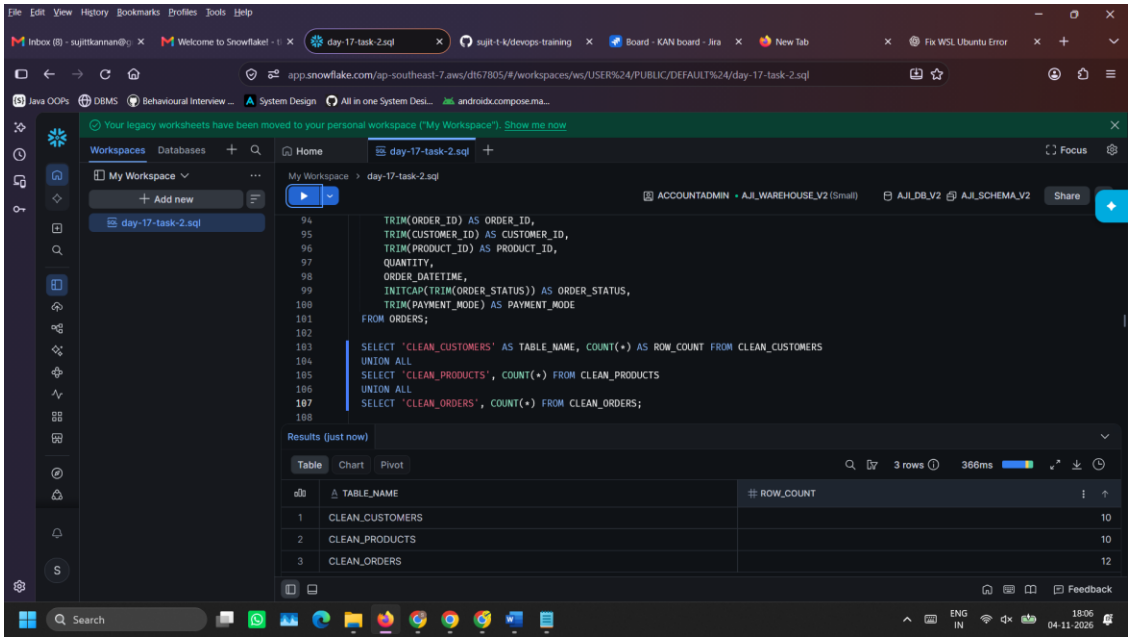
CREATE OR REPLACE TABLE CLEAN_PRODUCTS AS
SELECT
    TRIM(PRODUCT_ID) AS PRODUCT_ID,
    TRIM(PRODUCT_NAME) AS PRODUCT_NAME,
    INITCAP(TRIM(PRODUCT_CATEGORY)) AS PRODUCT_CATEGORY,
```

```
    UNIT_PRICE,
    STOCK_QUANTITY
FROM PRODUCTS;

CREATE OR REPLACE TABLE CLEAN_ORDERS AS
SELECT
    TRIM(ORDER_ID) AS ORDER_ID,
    TRIM(CUSTOMER_ID) AS CUSTOMER_ID,
    TRIM(PRODUCT_ID) AS PRODUCT_ID,
    QUANTITY,
    ORDER_DATETIME,
    INITCAP(TRIM(ORDER_STATUS)) AS ORDER_STATUS,
    TRIM(PAYMENT_MODE) AS PAYMENT_MODE
FROM ORDERS;
```



8.) Validate cleaned table entries (to ensure that new table holds data)

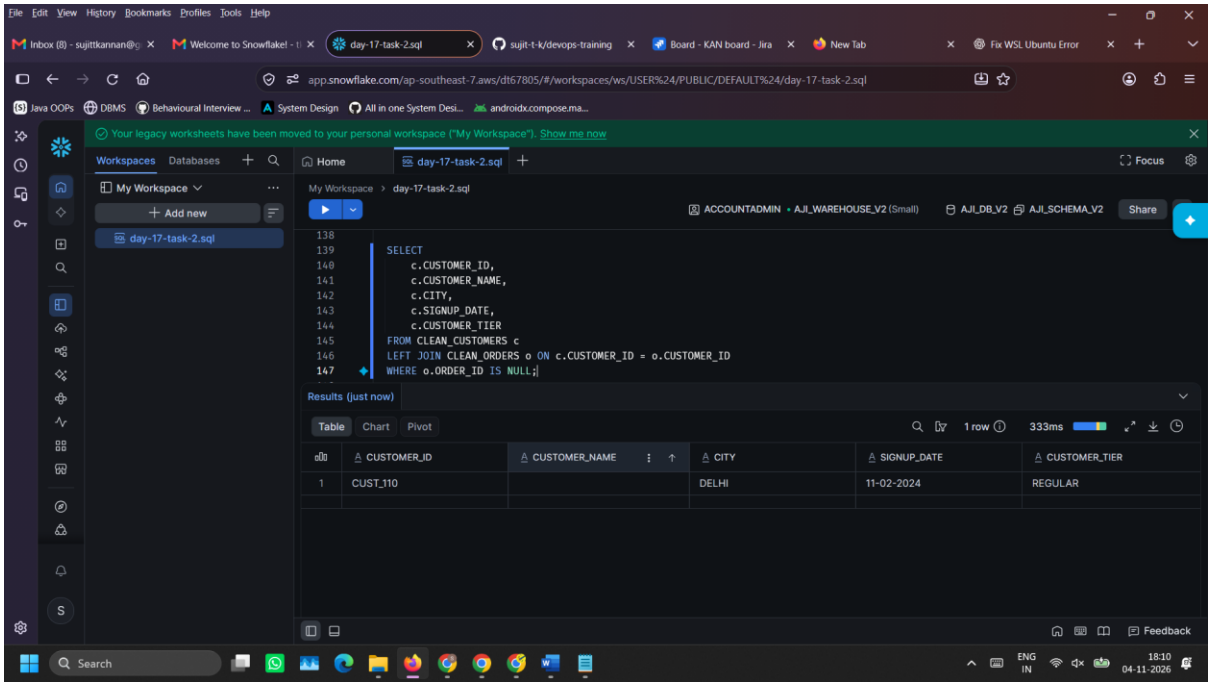


```
SELECT 'CLEAN_CUSTOMERS' AS TABLE_NAME, COUNT(*) AS ROW_COUNT FROM CLEAN_CUSTOMERS
UNION ALL
SELECT 'CLEAN_PRODUCTS', COUNT(*) FROM CLEAN_PRODUCTS
UNION ALL
SELECT 'CLEAN_ORDERS', COUNT(*) FROM CLEAN_ORDERS;
```

9.) Perform analytical queries

9.1) JOIN: Query to identify customers who have never made even a single order

```
SELECT
    c.CUSTOMER_ID,
    c.CUSTOMER_NAME,
    c.CITY,
    c.SIGNUP_DATE,
    c.CUSTOMER_TIER
FROM CLEAN_CUSTOMERS c
LEFT JOIN CLEAN_ORDERS o ON c.CUSTOMER_ID = o.CUSTOMER_ID
WHERE o.ORDER_ID IS NULL;
```

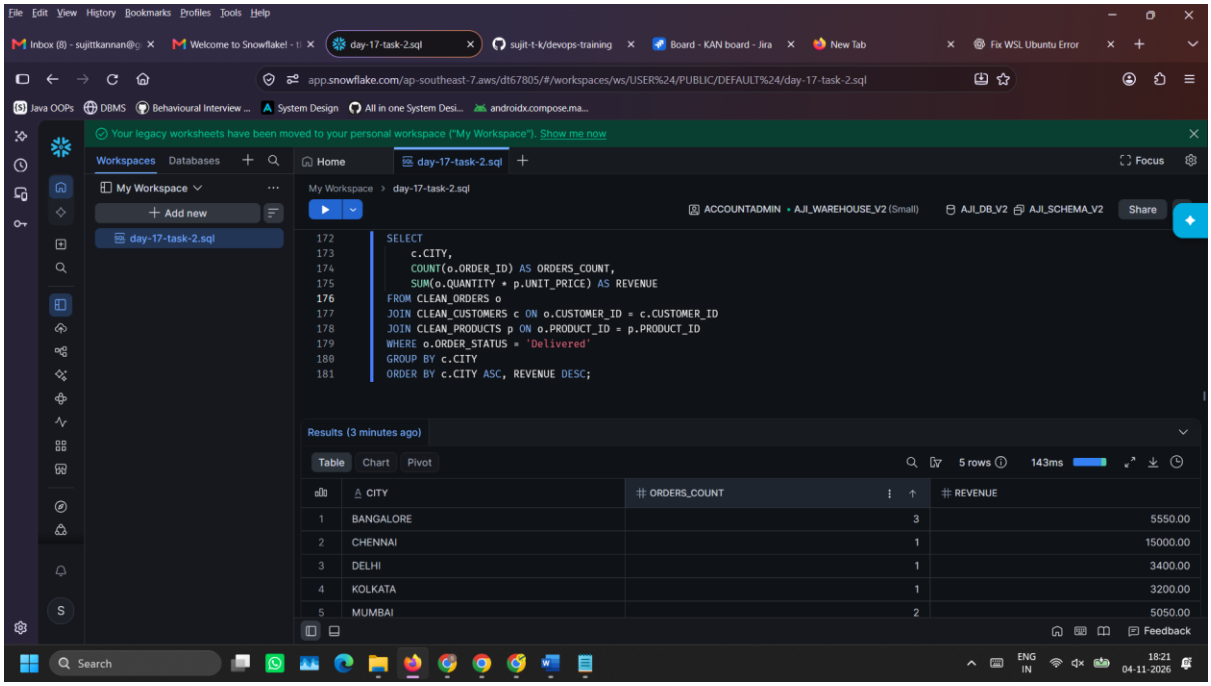


OBSERVATION FROM THE OUTPUT: The customer's name is empty but it does not get replaced with 'UNKNOWN CUSTOMER'

VERDICT: COALESCE() only REPLACES NULL values, it DOES NOT REPLACES non-null, empty values

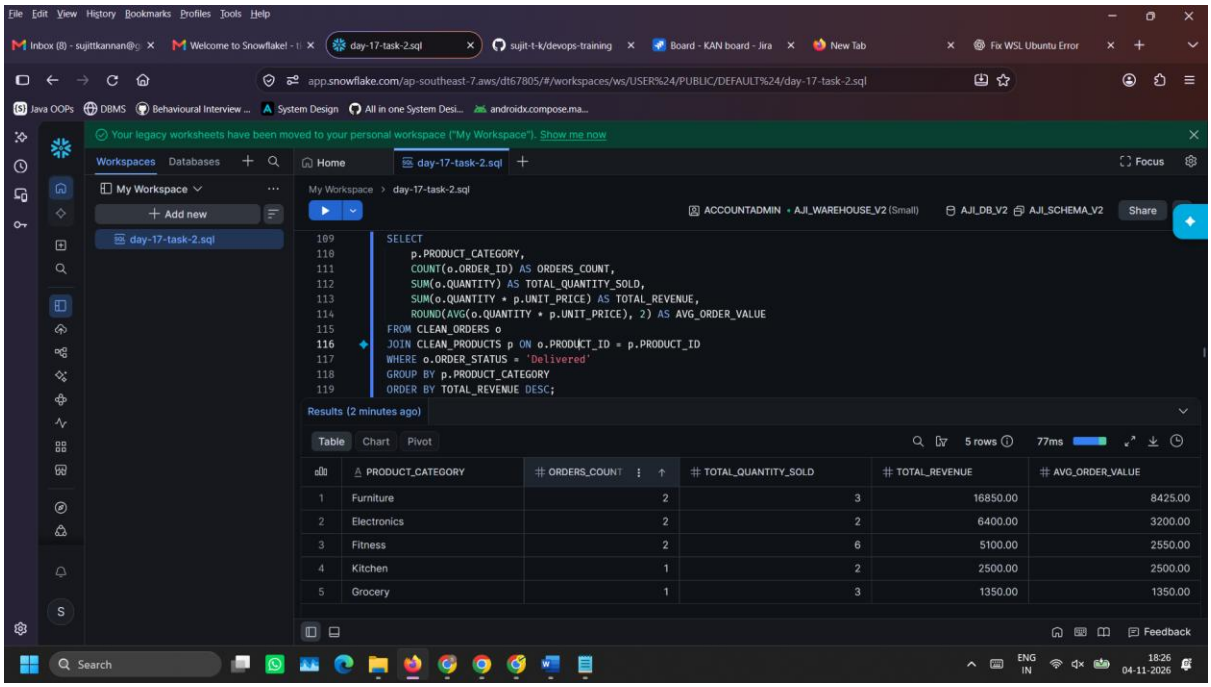
9.2) JOIN, GROUPING, AGGREGATION, FILTERING, SORTING: Revenue generated across each city

```
SELECT
    c.CITY,
    COUNT(o.ORDER_ID) AS ORDERS_COUNT,
    SUM(o.QUANTITY * p.UNIT_PRICE) AS REVENUE
FROM CLEAN_ORDERS o
JOIN CLEAN_CUSTOMERS c ON o.CUSTOMER_ID = c.CUSTOMER_ID
JOIN CLEAN_PRODUCTS p ON o.PRODUCT_ID = p.PRODUCT_ID
WHERE o.ORDER_STATUS = 'Delivered'
GROUP BY c.CITY
ORDER BY c.CITY ASC, REVENUE DESC;
```



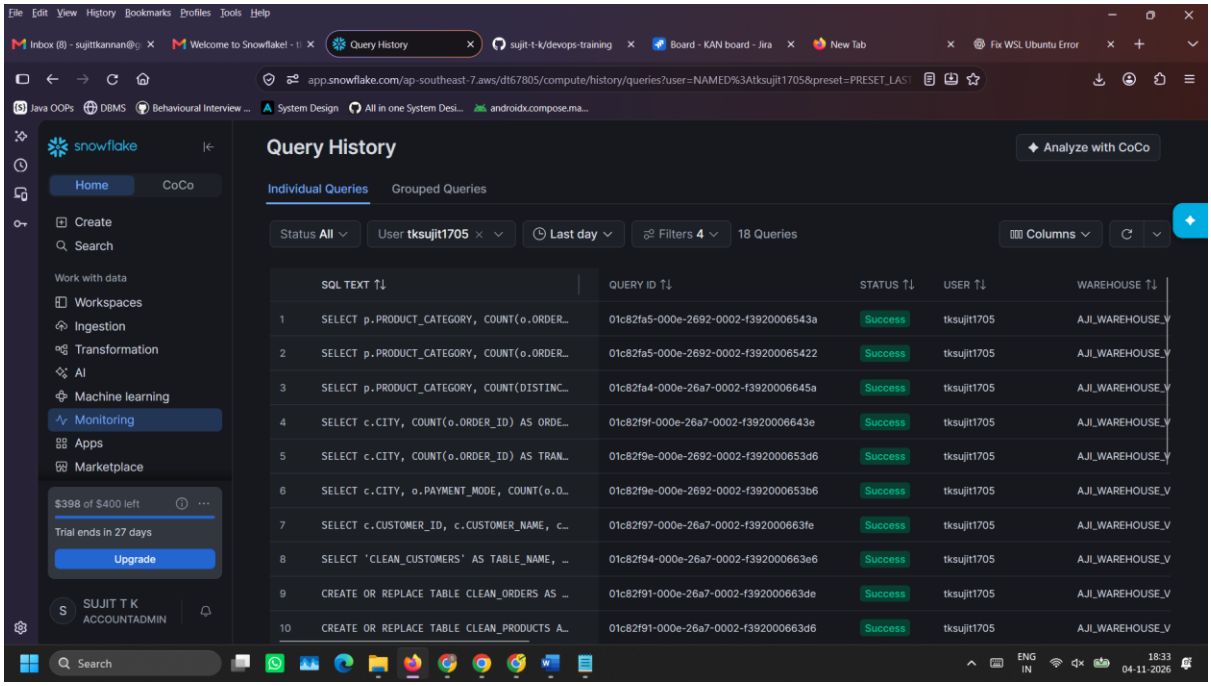
9.3) JOIN, GROUPING, AGGREGATION, SORTING: Product category-wise analysis on total orders and revenue

```
SELECT
    p.PRODUCT_CATEGORY,
    COUNT(o.ORDER_ID) AS ORDERS_COUNT,
    SUM(o.QUANTITY) AS TOTAL_QUANTITY_SOLD,
    SUM(o.QUANTITY * p.UNIT_PRICE) AS TOTAL_REVENUE,
    ROUND(AVG(o.QUANTITY * p.UNIT_PRICE), 2) AS AVG_ORDER_VALUE
FROM CLEAN_ORDERS o
JOIN CLEAN_PRODUCTS p ON o.PRODUCT_ID = p.PRODUCT_ID
WHERE o.ORDER_STATUS = 'Delivered'
GROUP BY p.PRODUCT_CATEGORY
ORDER BY TOTAL_REVENUE DESC;
```

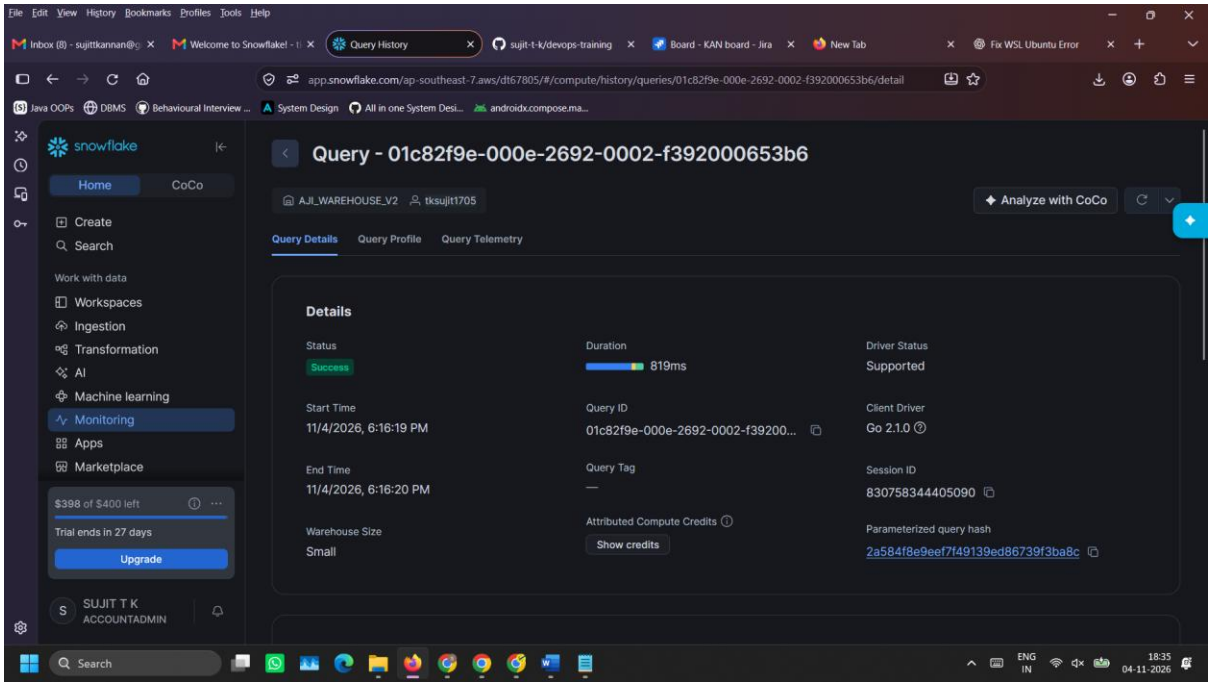


10.) Review query profile for optimization

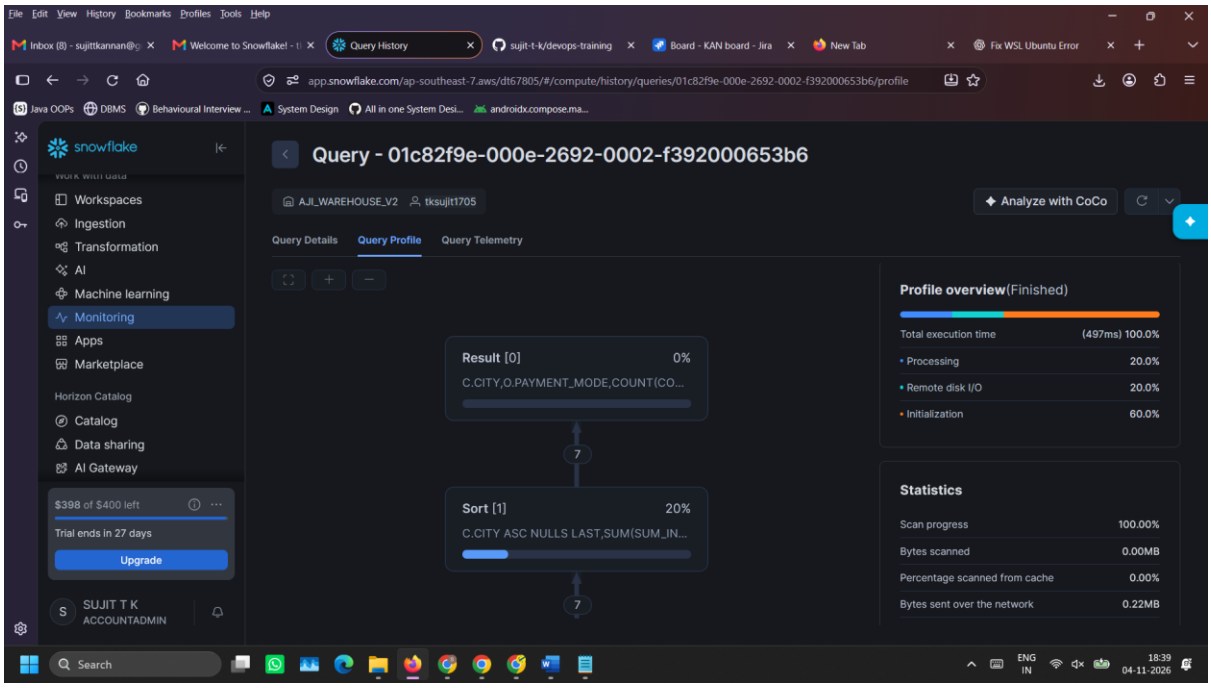
- On the left side bar, click on ‘Monitoring’ tab.
- Then go to ‘Query History’. It will show all the individual queries that were executed.



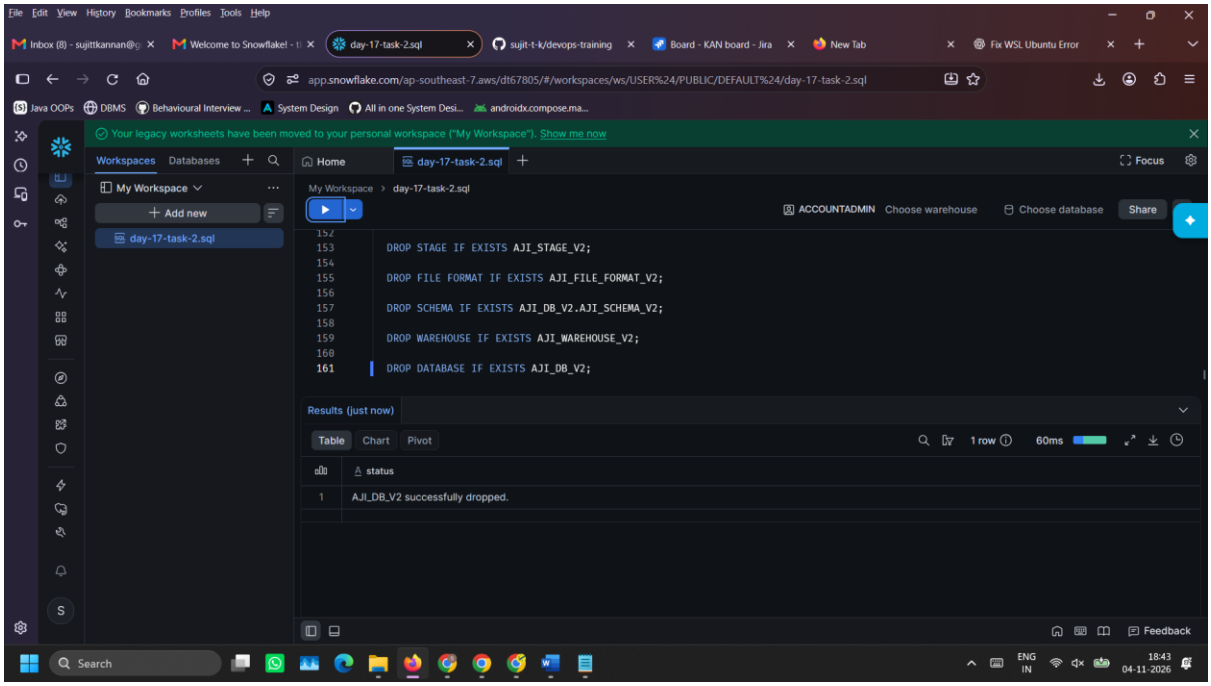
- Click on any of the queries to view the query information in detail.



Click on ‘Query Profile’ tab to view query insights. This helps in identifying which part of the SQL consumed more compute resources, so that the query can be optimized further.



11.) Delete Tables, warehouses, schema and database for cleanup to avoid additional charges




```
DROP TABLE IF EXISTS CLEAN_ORDERS;
DROP TABLE IF EXISTS CLEAN_PRODUCTS;
DROP TABLE IF EXISTS CLEAN_CUSTOMERS;
DROP TABLE IF EXISTS ORDERS;
DROP TABLE IF EXISTS PRODUCTS;
DROP TABLE IF EXISTS CUSTOMERS;

DROP STAGE IF EXISTS AJI_STAGE_V2;
```

```
DROP FILE FORMAT IF EXISTS AJI_FILE_FORMAT_V2;

DROP SCHEMA IF EXISTS AJI_DB_V2.AJI_SCHEMA_V2;

DROP WAREHOUSE IF EXISTS AJI_WAREHOUSE_ V2;

DROP DATABASE IF EXISTS AJI_DB_V2;
```